

Antonio Leblanc

Co-Founder & CTO — 1.5°C
AI Wildfire Platform

Rio de Janeiro, Brazil · French & Brazilian citizen
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190 cameras 150 HaaS on hardware we operate + 40 SaaS on third-party cameras	40M+ hectares Under wildfire monitoring, an area larger than Germany — camera, satellite, and geospatial data	98% of fires Spotted by our AI before any person, in production	9 AI agents Self-hosted Hermes fleet — 450+ merged PRs, >83% human acceptance, 10 repos
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EXPERIENCE

Co-Founder & CTO — 1.5°C, umgrauemeio Sep 2020 – Present
Rio de Janeiro, Brazil · Brazilian climate tech — AI for wildfire detection and response

- Architected Pantera, our wildfire platform, end to end — hybrid edge/cloud, combining 190 field cameras with multi-satellite and geospatial data for fire risk, early detection, fire spread, CO₂, and deforestation. **Tuned for recall**, since a missed fire costs more than a false alarm: in production, the AI spots **98% of fires** before a person does.
- Built the engineering, ML, and environmental science team from zero — **up to 15 people**, every one hired and developed by me; I still design architecture and review code with them.
- Deployed **9 AI agents** on self-hosted **Hermes** (model-agnostic by design) — an issue-to-PR pipeline with **450+ merged PRs** across 10 repos, every one human-reviewed (>**83%** acceptance, 2-PR WIP limit).
- Cut AWS costs **15%** through the engineering agent's infrastructure audits — it proposed the changes; the team reviewed and applied them.
- Took on **Customer Success, Support, and Marketing** at different points and led HR for two years, handing each area to a dedicated lead.
- Technical lead for **Suzano, the world's largest pulp producer** and our largest customer, won in a competitive bid — built the analytics module promised in the deal, working closely with their team, then climbed camera towers on site to get the network running.

Software Engineer Apr – Aug 2020
Machine Learning Intern Nov 2019 – Mar 2020

Earlier experience 2018 – 2019

- Radix Engenharia e Software** — Data Science Intern, Rio de Janeiro: genetic optimization algorithm that **cut energy consumption 20%** in industrial processes; ML and digital-twin models for predictive maintenance.
- Amazon** — Operations Management Intern, Orléans, France: **managed a team of 20** and supply chain optimization in a fulfillment center processing 400K units/day.

OPEN SOURCE

ForeFire — open-source wildfire spread simulator in C++, developed at CNRS / Université de Corse 2022 – Present

- Contributor since 2022, maintainer since 2025 — reviewing and merging PRs, cutting releases, Dockerization, CI/CD, documentation.
- Co-author on the project's peer-reviewed **JOSS paper** (DOI 10.21105/joss.08680).

EDUCATION

General Engineering École Centrale de Lyon, France	2016 – 2018
Mechanical Engineering Polytechnic School of UFRJ, Brazil · CAPES scholarship for academic excellence	2014 – 2019
Baccalauréat Scientifique Lycée Molière, Rio de Janeiro, Brazil · Highest Honors	2013

SKILLS & LANGUAGES

LANGUAGES	Native: Portuguese, French · Fluent: English · Professional: Spanish
STACK	Python, FastAPI, TypeScript/Angular, AWS, Docker, PostGIS, PyTorch
ML/AI	Deep learning, geospatial ML in production, LLMs and agentic systems
LEADERSHIP	Agile/OKRs, cross-functional teams across engineering, ML, and environmental science
ALSO	Marathon (3:23) · 10 years of formal piano; self-taught saxophone, guitar